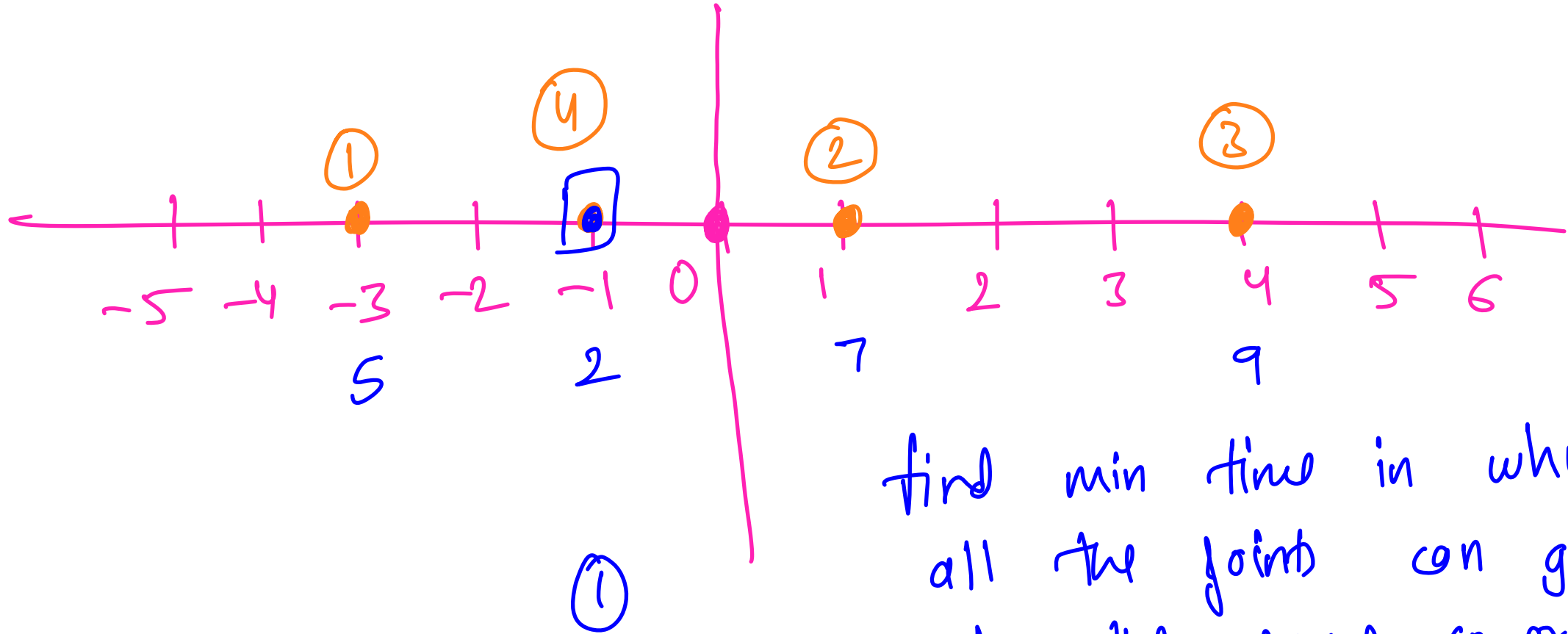


Advanced Binary Search 🍌

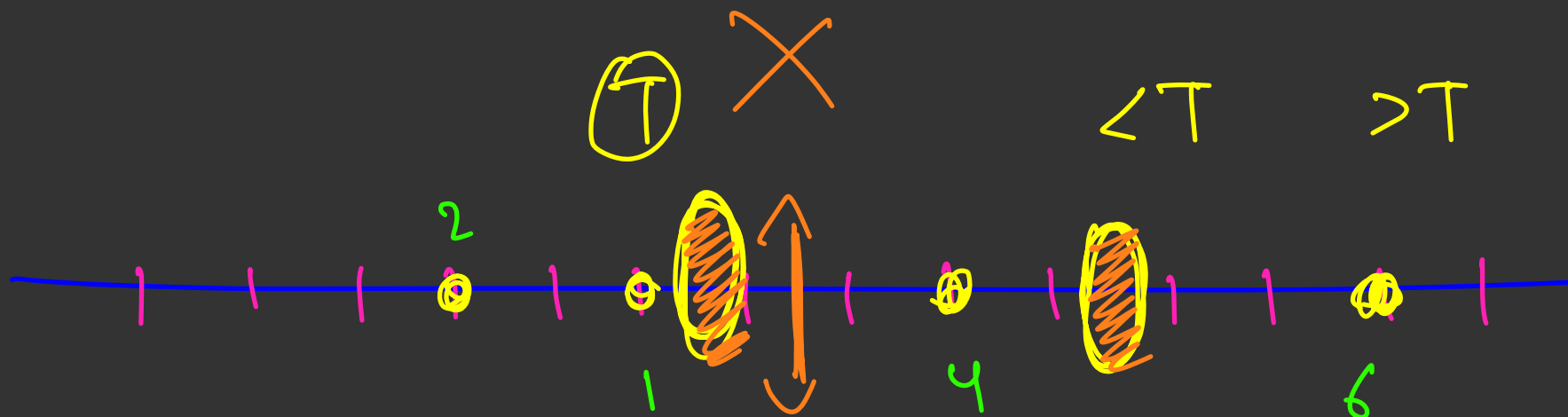
Problem Solving 2

Problem 1: Get Together

n points



find min time in which
all the points can gather
at the same co-ordinate



$$\max_{i \rightarrow 0 \text{ to } n-1} \left(\frac{|X - x_i|}{v_i} \right) \quad \frac{-10^9 \text{ to } 10^9}{\sqrt{10^6}}$$

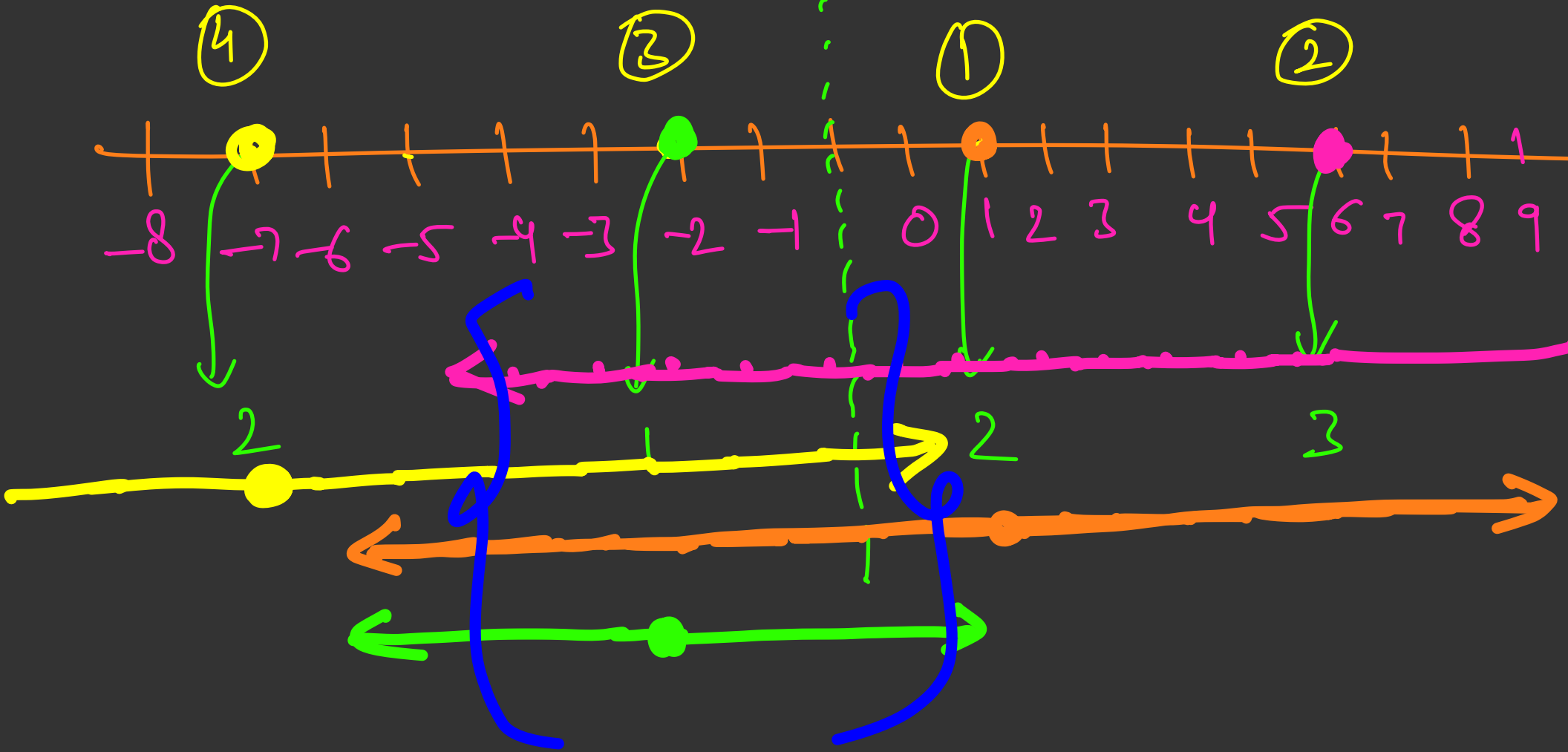
$$f(x) = \max_{i \rightarrow 0 \text{ to } n-1} \left\{ \frac{|x - x_i|}{v_i} \right\} \quad 2 \times 10^9 \times 10^6 \rightarrow \underline{\underline{2 \times 10^{15}}}$$

Binary search on time.

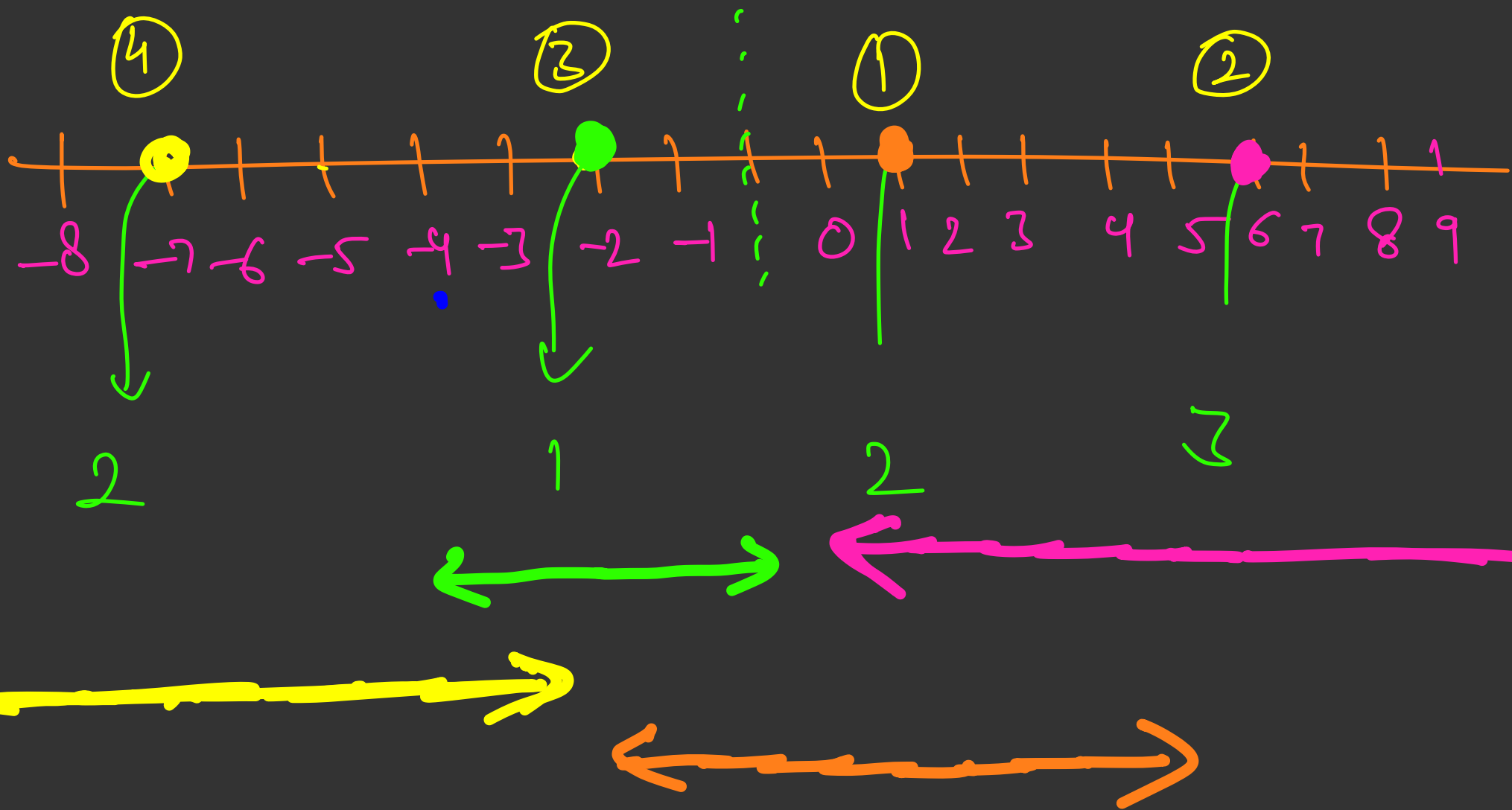
$f(t) = \begin{cases} T & \text{if all points can gather together in } \leq t \text{ time} \\ f & \text{otherwise} \end{cases}$

f f f f f f f f T T T T T T T T T T

time = 4s



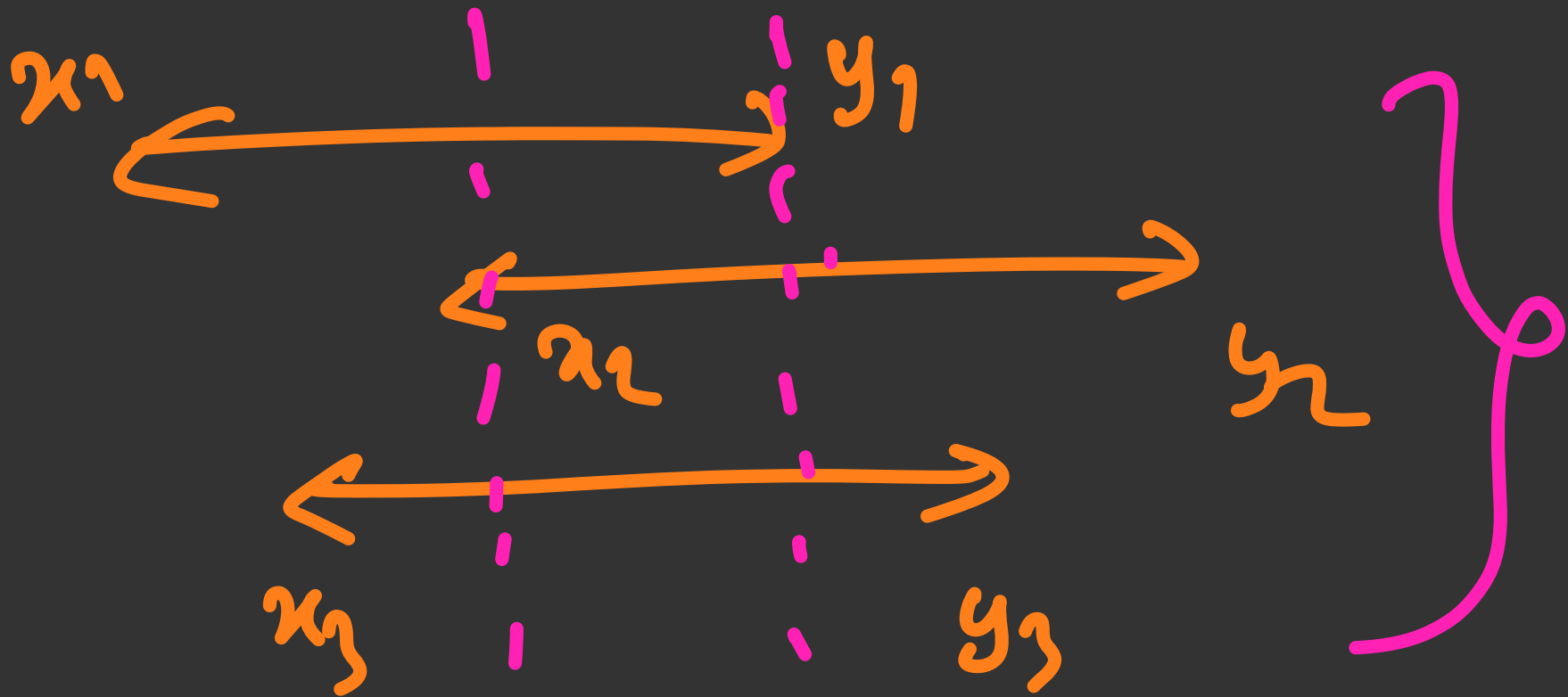
time = 2





$$\max(x_1, x_2) \leq \min(y_1, y_2)$$

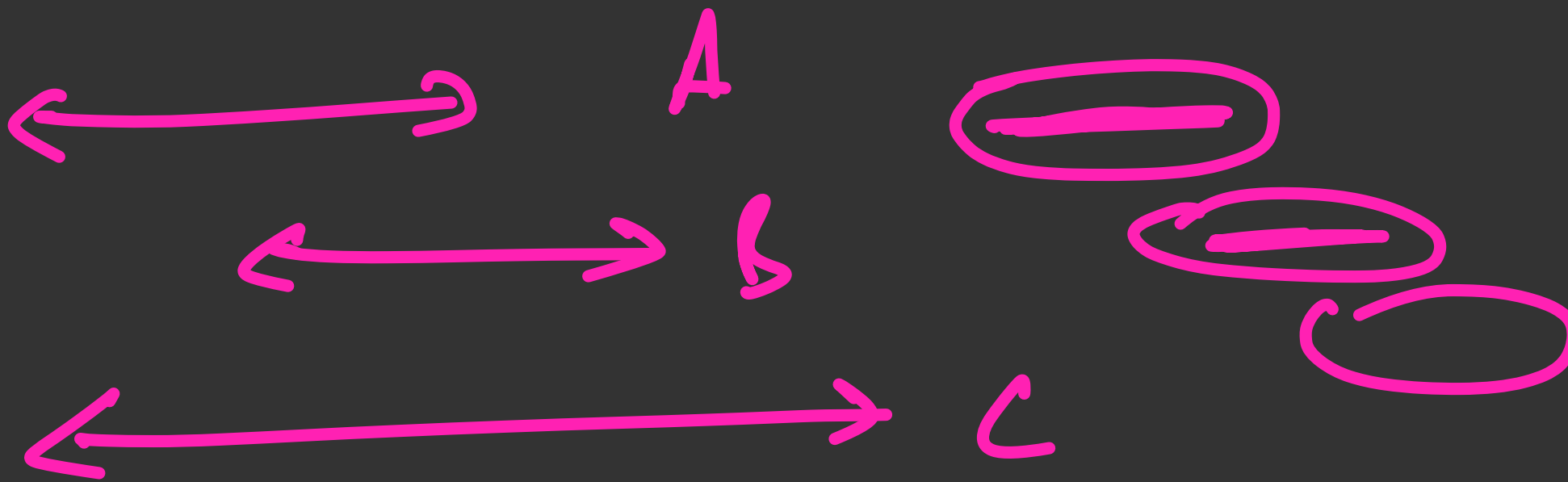




$$\max(x_1, x_2, x_3) \leq \min(y_1, y_2, y_3)$$

$$\leftarrow \bullet A \quad \begin{matrix} V^T \\ \downarrow \end{matrix}$$

$$[A - VT \leftarrow A + VT]$$



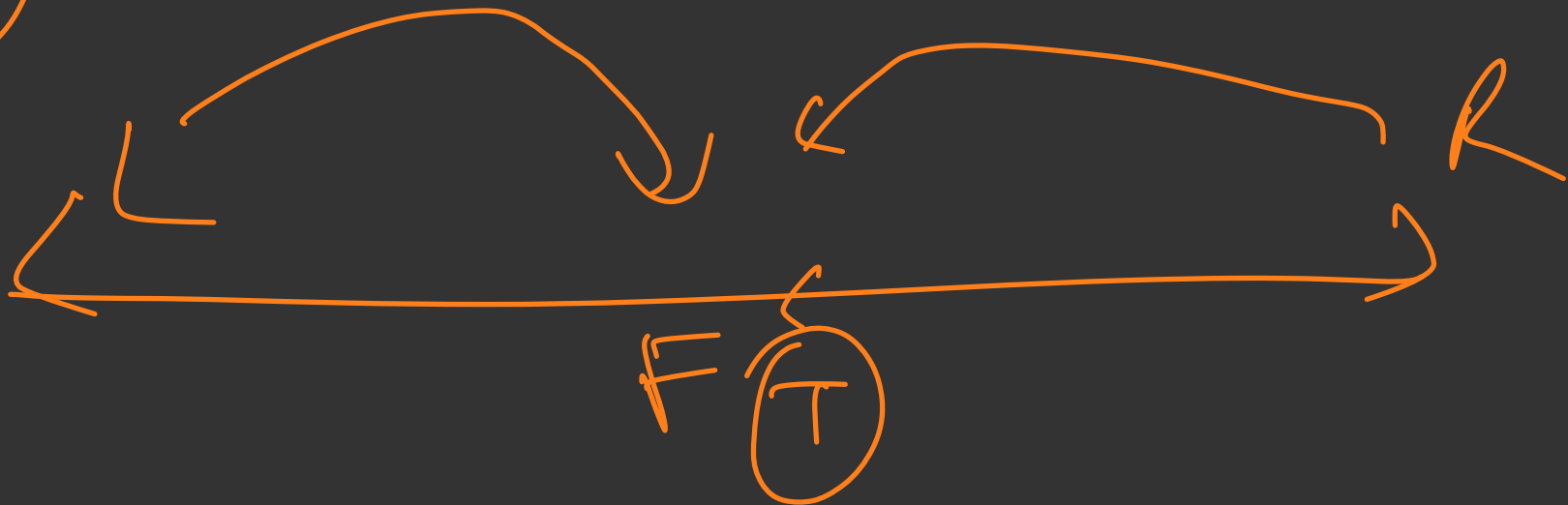
intersection (A, B, C)

intersection (intersection (A, B), C)



Ans

f f f f f T T T T T T T



Method ①

fffff(T)TTTTTTT

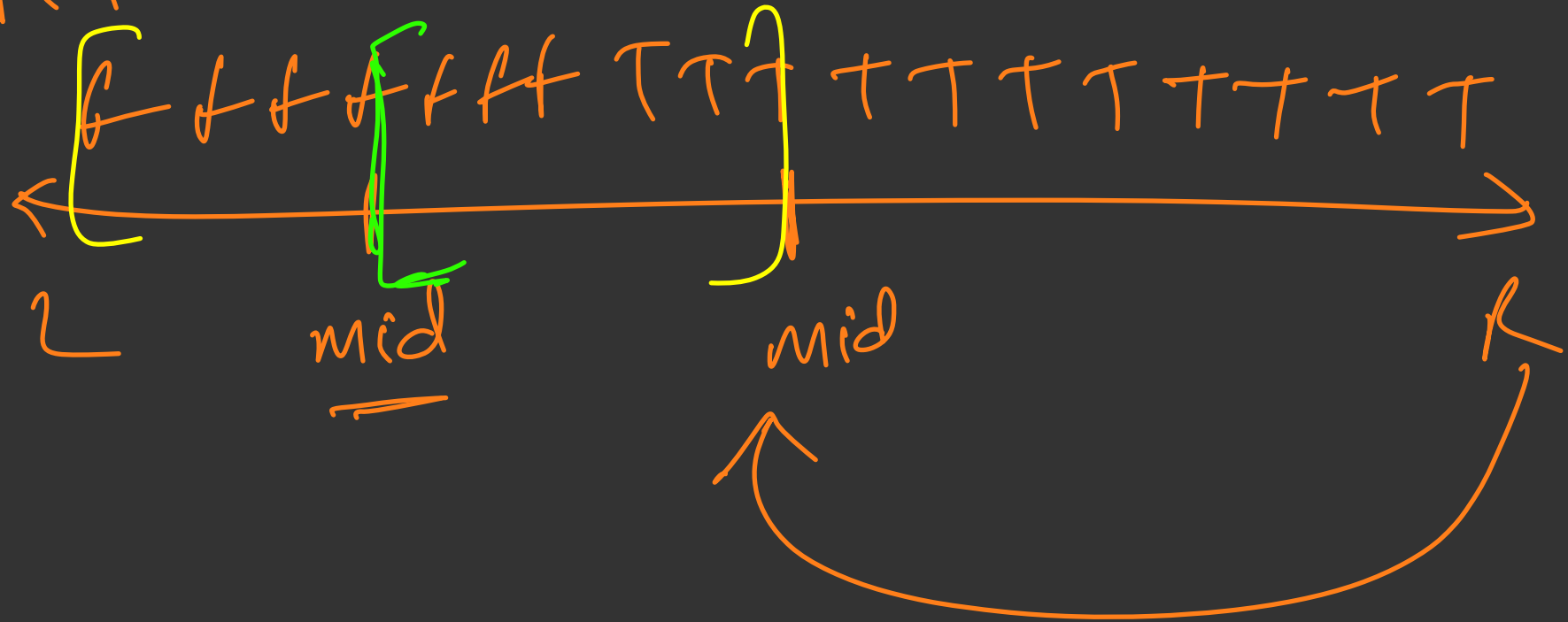


maintain ans variable

$$R = \text{mid} - \text{error}$$

$$L = \text{mid} + \text{error}$$

Method ②



Problem 2: Tracking Segments

[CP-31 1600: P5]